

Connections for composite concrete slab and LVL flooring systems

Bruce L. Deam · Massimo Fragiaco ·
Andrew H. Buchanan

Received: 16 January 2007 / Accepted: 9 May 2007 / Published online: 5 June 2007
© RILEM 2007

Abstract Composite concrete slab and timber flooring systems are commonly used in many parts of the world to exploit the high strength-to-weight ratio of timber and the good acoustic separation provided by concrete floor slabs. This paper describes the results of an experimental programme that investigated the suitability of a range of connectors to transfer shear between a concrete slab and a laminated veneer lumber (LVL) beam. Shear tests on reduced scale specimens were performed with the aim of comparing the strength, stiffness, and post-peak performance of different connectors such as round and rectangular concrete plugs with and without screw and steel pipe reinforcement, proprietary (SFS) screws, coach screws with different diameters, sheet brace anchors,

and framing brackets. The rectangular concrete plug reinforced with a coach screw was found to provide the greatest stiffness and strength, as well as favourable post-peak behaviour. Such a system can be used for cost-effective composite floor systems due to its efficient cost-to-capacity ratio, which reduces the number of connectors needed along the beam axis to achieve the composite action.

Keywords Connector · LVL · Structural engineering · Timber–concrete composite beams · Wood

1 Introduction

Composite concrete and timber (CCT) beams represent a structural technique widely used for both strength and stiffness upgrading of existing and new floors. This technique consists of connecting an existing or new timber beam to a concrete slab cast above the timber beam, using a suitable connection system [1]. Steel mesh is usually cast into the slab to resist potential tensile stresses due to bending, and to reduce the concrete crack width. CCT beams provide many benefits compared with traditional timber floors, including greater strength and stiffness, less susceptibility to vibration, improved seismic and fire resistance, better acoustic separation and thermal mass. The lower weight than reinforced concrete floors also imposes less load on the foundations and

B. L. Deam
Department of Civil Engineering,
University of Canterbury, Private Bag 4800,
Christchurch 8140, New Zealand
e-mail: bruce.deam@canterbury.ac.nz

M. Fragiaco (✉)
Department of Architecture and Planning, Faculty of
Architecture, University of Sassari, Palazzo del Pou Salit,
Piazza Duomo 6, 07041 Alghero, Italy
e-mail: fragiaco@uniss.it

A. H. Buchanan
Department of Civil Engineering,
University of Canterbury, Private Bag 4800,
Christchurch 8140, New Zealand
e-mail: andy.buchanan@canterbury.ac.nz



in earthquake-prone regions reduces the lateral strength required within the structural system.

The performance of the CCT beam elements largely depends on the choice of the connection system. Stiff and strong shear connections are required to provide optimal structural efficiency. By preventing relative slip between the bottom fibre of the concrete slab and the top fibre of the timber beam, a high degree of composite action can be achieved, where the concrete slab acts as a flange of T-shaped cross-section and the timber beam acts as a web. This fully exploits the properties of both materials, with the concrete mainly in compression and the timber in both tension and bending. The connection system should preferably provide ductility within the CCT element, since both timber and concrete are naturally quite brittle (in tension and compression respectively) [2, 3]. In contrast, the connection system needs to be inexpensive to manufacture and install in order to make CCT beams competitive with other construction systems. Many researchers have investigated this aspect and several types of connectors have been proposed [1, 4–17].

The connection system can be either discrete, such as when it is fabricated using a number of discrete mechanical fasteners, or it can be a continuous element along the timber beam. Common discrete systems include coach screws, specific proprietary screws such as SFS screws [18], nail plates, dowels, rebars, stud connectors such as the “Tecnaria connector” screwed into the timber [5], plugs with different shapes (with or without mechanical reinforcement), and notched details. Examples of continuous systems include steel lattices glued and milled into the timber, punched metal plates glued and milled into the timber, glued-in steel meshes such as the HBV system [9], and punched steel profiles screwed into the timber. Glued connectors have the advantage of being stiffer and stronger than their mechanical counterparts, however the use of epoxy resin may significantly increase the cost and difficulty of construction.

The timber industry is currently looking for new applications for timber use in multi-storey buildings. CCT floor systems have not been used in New Zealand or Australia because they have not been able to match the long-span floors (8–10 m) possible with other materials, but they may be a viable alternative, provided the system is both efficient and cost effective.

Laminated Veneer Lumber (LVL) can be used to provide both better reliability and better strength and

stiffness than either solid or glulam timber. The LVL production process minimises the influence of wood defects such as knots and grain deviation by randomly distributing these through the member volume. The use of 3 mm thick veneers within LVL ensures that cracks which develop at these defects are too small to propagate into the adjacent veneers. The LVL is also less susceptible to splitting than sawn timber or glulam.

The paper reports the outcomes of a series of experimental tests performed at the University of Canterbury, which investigated the feasibility of composite concrete slab and LVL flooring systems. The goal of this pilot study was to identify the most cost effective connection systems. The performance of different connection systems is evaluated through experimental shear tests, in order to find the best compromise between cost and structural efficiency. The connector spacing is then calculated for an 8-m span CCT office floor in order to explore the economic feasibility of the composite solution.

2 Properties of the materials

An experimental programme was carried out to investigate whether composite concrete slab and LVL beams are a viable and convenient alternative to the more commonly used reinforced concrete or traditional timber floors. The connection system needs to provide the best possible compromise between structural efficiency and the cost of construction. For this purpose, a range of very different connection systems were tested to investigate their performance, in terms of stiffness, strength and ductility achievable. The tests were performed on small specimens comprising a concrete slab attached to the top of a short length of LVL. These were arranged so the connection was mainly subjected to shear. The properties of the component materials (concrete, LVL and the connections) are described in the following sections.

2.1 Concrete

A single batch of concrete was prepared in the laboratory using the mix design given in Table 1. A piece of 6 mm diameter, 150 × 150 mm steel mesh reinforcement was placed at mid-depth of the concrete

Table 1 Mix design of the concrete

Component	Amount per cubic meter
Holcim GP Cement (equivalent to CEM I 53.5 R cement)	285 kg
Water	170 kg
Stone—13 mm Greywacke sandstone	1,000 kg
Sand—4.75 mm Greywacke sandstone	940 kg
Water reducer (Modified ligno sulphonate)	850 ml
Air entraining agents (Synthetic chemical blend)	180 ml

slab. Once poured onto the specimens, the concrete was allowed to cure by moistening for 6 days and then covering it with wet sacks for a further 13 days. Nine 100×200 mm cylinders (three per cast) were also taken and tested in compression 28 days after the concrete was poured. The cylinders were tested in accordance with NZS 3112 [19], giving an average cylindrical compression strength of 30.4 MPa and average density of $2,293 \text{ kg/m}^3$, with standard deviations of 1.3 MPa and 8 kg/m^3 , respectively.

2.2 Timber

The timber used in the tests was LVL, which is laminated from rotary peeled, parallel-arranged veneers graded for strength, staggered, lapped and glued with structural adhesive [20]. The manufacturing process, which minimizes the size of the defects such as knots and randomises and their location, leads to a product characterized by reduced variability and improved mechanical properties with respect to sawn timber. The properties of the LVL used for the specimens (marketed under the trade name Hyspan and supplied by Carter Holt Harvey Ltd) are given in Table 2 [21]. The Young's modulus, measured using a resonance-based method [22], was found to be $E = 14.4 \text{ GPa}$, slightly higher than the values suggested by the manufacturer [21]. The average density of the LVL was 603 kg/m^3 . The $105 \times 140 \times 430$ mm beam segments used to construct the specimens were all cut from one 105×210 mm billet of LVL 2 m long.

2.3 Connection systems

A range of connection systems were proposed, on the basis of structural efficiency, cost of the connector

Table 2 Mechanical properties of LVL marketed as Hyspan [21]

Mechanical property	Value (MPa)
Modulus of elasticity	13,200
Modulus of rigidity	660
Characteristic stress in bending	42
Charact. stress in tension parallel to grain	27
Charact. stress in compr. parallel to grain	34
Characteristic stress in longitudinal shear	4.5
Char. stress in compr. perpendicular to grain	12

and labour cost. The details of those that were tested are summarised in Tables 3 and 4. The concrete plug, both with [23, 24] and without reinforcement, appeared to provide a simple yet effective solution, so four different combinations of plug geometries (circular or rectangular) and type of reinforcement (coach screw or steel pipe) were tested. The concrete plug dimensions were selected to provide the greatest bearing area, while minimising the depth of the plug protruding into the timber beam [25]. A range of simple mechanical connectors were also tested that did not require the LVL to be notched, such as the Pryda sheet brace anchor, the Pryda bracket [26], coach screws and SFS screws [18] (with a range of fastener orientations). The Pryda connectors were screwed into the LVL beam, while the SFS screws were prebored at 45° to the LVL-concrete interface.

Only a small number of specimens were tested for each type of connection (two for the notched connections, one for the mechanical systems). Such a reduced number cannot allow drawing final conclusions on the values of strength and shear moduli, however, the performances of the different connections can be compared to each other in order to achieve the goal of this pilot study which was to identify the most cost effective systems. The reliability of LVL [20], in fact, markedly reduces the scatter of the connection properties when compared with sawn timber, and enables qualitative conclusions even with few specimens.

3 Experimental set-up

Shear tests can be conducted with either asymmetrical or symmetrical specimens as shown in Fig. 1. In this experimental programme, the asymmetrical con-



Table 3 Characteristics of the plug connectors tested

Number	Picture of the connector	Description
1 & 2		type: round plug plug: 48.5 mm diameter 20 mm depth
3 & 4		type: round plug with screw plug: 48.5 mm diameter 20 mm depth screw: 12 mm diameter 9 mm predrilled 150 mm length 100 mm embedment length
5 & 6		type: reinforced plug plug: 48.5 mm diameter 20 mm depth pipe: 40 mm length
7 & 8		type: rectangular plug with screw plug: 50 mm length 16.5 mm depth screw: 12 mm diameter 9 mm predrilled 150 mm length 103 mm embedment length

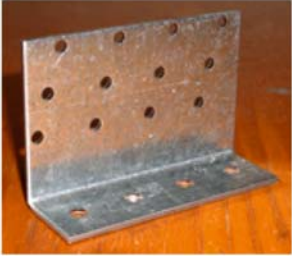
figuration was chosen, with the dimensions shown in Fig. 2. This type of configuration has the advantage of requiring the cast of a concrete slab on only one side of the LVL beam, with benefits in terms of time and cost. The connectors were all located in the centre of the 430 mm long LVL segment. Specimen 17 had a 17 mm thick plywood sheet between the LVL and concrete to represent permanent formwork, but the concrete and LVL dimensions were otherwise the same as those in Fig. 2.

The experimental arrangement is shown in Figs. 3 and 4. The specimens were inverted to simplify the test arrangement and placed onto a plywood sheet above a steel base plate. The steel plate provided a flat surface on top of the strong floor and the plywood plate accommodated irregularities within the concrete surface. A steel angle section (labelled fixed base) limited horizontal movement of the concrete slab.

Horizontal load was applied using a 120 kN hydraulic actuator operated with a hand pump. Rollers were positioned both behind and in front of the specimen to support the weight of the loading apparatus and a coaxial load cell. A third roller was placed at the end of the LVL beam to limit uplift of the specimen due to vertical eccentricity between the actuator and the reaction on the fixed base. The uplift force was measured using a second load cell.

The horizontal load was monotonically increased to failure in all the tests [27]. Special care was used to ensure a tight contact between the loading plate and the specimen before the beginning of the test. Although the actual slip required in timber–concrete composite beam connections is small, the specimens were all tested to failure at large deflection in order to collect information on the post-peak behaviour, ductility and type of failure. The relative slip between

Table 4 Characteristics of the mechanical connectors tested

Number	Picture of the connector	Description
9		Pryda steel brace anchor with rods placed parallel to the beam axis
10		Pryda steel brace anchor placed parallel to the beam axis
11		Pryda steel brace anchor placed at a 45° angle to the beam axis
12	 4 × $\phi 5,5 \times 45$ mm	Pryda steel brace anchor placed perpendicular to the beam axis
13		Coach screw: 12 mm diameter 9 mm predrilled 165 mm length 115 mm embedment length
14		Coach screw: 16 mm diameter 12.5 mm predrilled 155 mm length 105 mm embedment length
15		Pryda framing bracket FB47/76'
16	 6 × $\phi 3 \times 40$ mm	Inverted Pryda framing bracket FB47/76'
17		Two SFS screws $\phi 7,5 \times 100$ mm inclined at 45°

the timber beam and the concrete slab was monitored using linear potentiometers placed on each side of the LVL at the centre of each specimen (Fig. 5). Other potentiometers were used to monitor the vertical movement of the LVL relative to the concrete.

4 Results of the shear tests

All specimens were tested to failure. The relationship between shear force and relative slip is shown in Fig. 6 for all of the connection systems in the range of interest. Table 5 summarizes the specimen strength ($F_{\max}$), the slip (v) corresponding to the maximum strength, the secant shear moduli at 40% ($k_{0,4}$), 60%

($k_{0,6}$) and 80% ($k_{0,8}$) of the strength [28], and the type of failure (ductile or brittle). The strength $F_{\max}$ is defined as the greatest shear force measured during the test before a slip of 15 mm occurs [28]. This strength is used to design the connection in timber–concrete composite beams at Ultimate Limit State (ULS). The shear moduli $k_{0,4}$, $k_{0,6}$ and $k_{0,8}$ are used for both Serviceability Limit State (SLS) and ULS verifications as specified in [29, 30], and for the prediction of the average collapse strength of a timber–concrete composite beam [2].

The failure was defined as brittle when the connection could not withstand a relative slip of 10 mm without a drop in strength larger than 20% of the peak value. All others were defined as ductile.

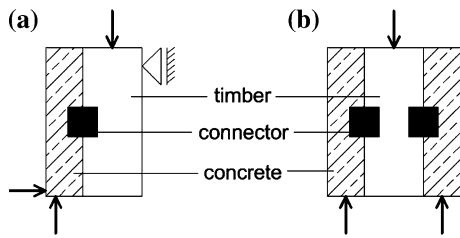


Fig. 1 Shear test configurations: (a) asymmetrical, (b) symmetrical

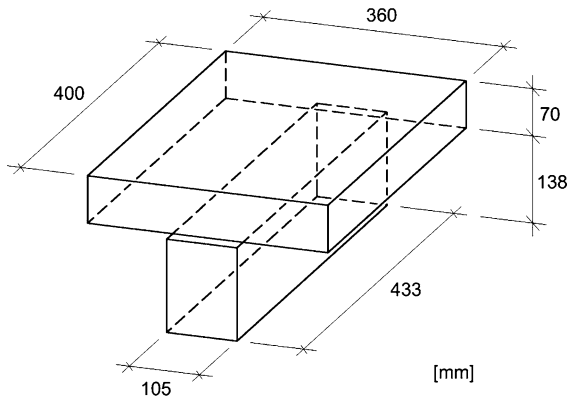


Fig. 2 Geometry of the specimen (dimensions in mm)

This definition of ductility is based on the evidence that the slip demand in composite beams is generally lower than 10 mm. Drops in strength larger than 20% are regarded as failure for brittle materials [31] and, if occurring in connections for slips lower than 10 mm, may cause a brittle shear failure of the composite beam.

4.1 Concrete plug specimens

Specimens 1 and 2 (with round unreinforced concrete plugs) displayed no measurable movement until the load was about 2 kN. The LVL rotated about a vertical

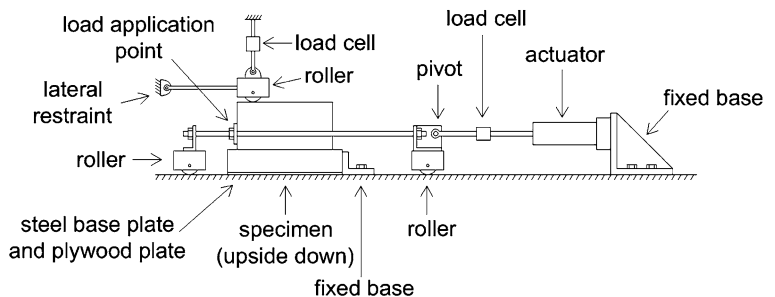
axis relative to the concrete, possibly due to asymmetric loading or asymmetric material properties in the concrete plug. After the load reached about 4 kN, the LVL translated with minimal rotation throughout the remainder of the test. The strain energy stored in the hydraulic hoses caused rapid slip increase after the concrete plugs fractured in shear at a load of 13.2 kN (Fig. 7a). The load dropped too rapidly for the data acquisition to capture the load or slip accurately.

Friction between the LVL and the concrete and between the two fractured plug surfaces is thought to have contributed most of the remaining bearing capacity after the maximum load. The joint was classed as brittle. The LVL began to lift off the concrete at the free end when the slip was about 2 mm displacement.

The stiffness, strength and ductility of the specimens with the round concrete plug were all increased by the addition of a coach screw (Specimens 3 and 4). The coach screw probably provided about half of the shear strength when the slip was small (Specimen 4 reached 31.4 kN, more than twice the strength of the unreinforced specimens), and for greater slip, the coach screw appeared to hold the fractured surfaces of the concrete plug together so the strength did not drop as rapidly as the slip increased. The screw eventually fractured in Specimen 4, causing the strength to drop rapidly. The screw in Specimen 3 had developed a double curvature shape by the end of the test and did not fracture (see Fig. 7b).

Reinforcement of the round concrete plug with a steel pipe (Specimens 5 and 6) produced a similar initial response as the specimens with round concrete plug and coach screw. However, after reaching the maximum load of 32.6 kN, both the concrete and LVL were crushed within the compression zones. The pipe tilted after the concrete fractured, forcing the LVL to lift off the concrete (see Fig. 7c). The

Fig. 3 Experimental set-up of the shear test



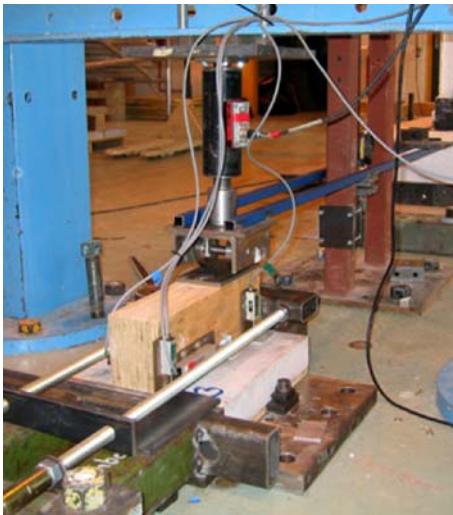


Fig. 4 View of the specimen under testing

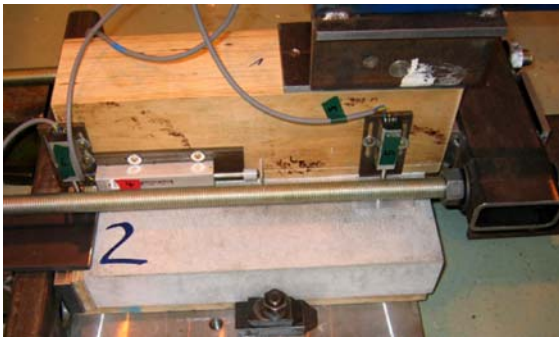


Fig. 5 Location of the potentiometers on the specimen

separation was accentuated at the right end of the specimen because it was not free to lift at the left end.

The rectangular plug reinforced with a coach screw (Specimens 7 and 8) failed in a similar manner

to the matching specimen with round concrete plug and screw (Fig. 7d). The significantly greater strength (54.9 kN) and stiffness were attributed to both the larger plug shear area and the larger bearing surface area.

The stiffness and strength of this form of connection is therefore governed by the bearing surface area of the concrete plug, the shear area of the plug and both the flexural and the tensile strengths of the screw. The presence (or absence) of a screw dictates the post-peak behaviour and therefore both the ductility and the strength degradation.

4.2 Steel brace anchors and framing brackets

The responses of Specimens 9 and 10, with steel brace anchors arranged parallel to the LVL beam, were both similar, indicating that concrete was adequately bonded to the plate without the additional horizontal rods provided in Specimen 9. Damage in both specimens was confined to the screwed connections between the steel bracket and the LVL.

The orientation of the brace anchors (Specimens 11 and 12 with the plates at 45 and 90° to the beam axis respectively) had no significant effect on the performance as they all had the same number of fasteners embedded in the LVL.

Similar responses were observed with the Pryda framing brackets (Specimens 15 and 16), with all of the slip being concentrated in the nailed connection between the steel bracket and the LVL. However, the ductility was markedly larger for the nailed connection than it was for the screwed connections. Again, the orientation of the fully embedded brackets in the concrete had no effect. The spiral nails were completely extracted from the LVL at the end of

Fig. 6 Shear force-relative slip curves for the different connection systems tested

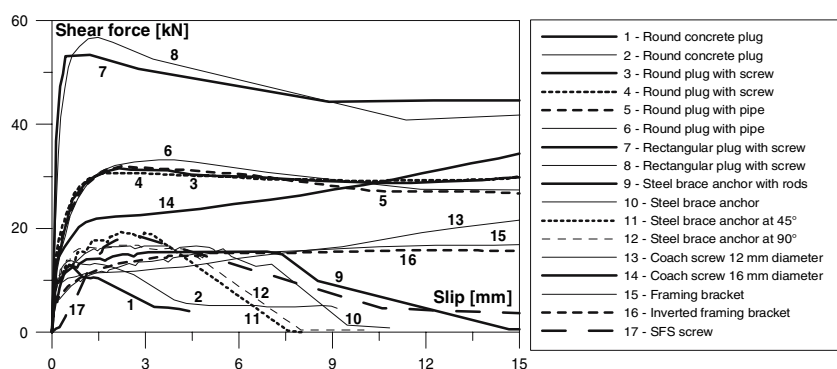


Table 5 Strength, slip at the attainment of the strength, secant shear moduli, and type of failure for the tested connection systems (average values for pairs of specimens)

Type of connector	Spec.	$F_{\max}$ (kN)	v (mm)	$k_{0.4}$ (kN/mm)	$k_{0.6}$ (kN/mm)	$k_{0.8}$ (kN/mm)	Type of failure
Round concrete plug	1 & 2	13.2	0.62	83.1	42.6	36.9	Brittle
Round plug with screw	3 & 4	31.4	1.89	105.9	56.3	39.1	Ductile
Round plug with pipe	5 & 6	32.6	3.09	66.6	42.5	29.7	Ductile
Rectangular plug with screw	7 & 8	54.9	1.35	297.0	197.3	148.5	Ductile
Brace anchor with rods	9	15.3	6.89	156.9	125.8	20.6	Brittle
Brace anchor	10	16.6	2.51	170.3	158.1	18.8	Brittle
Brace anchor at 45°	11	19.3	2.96	77.2	30.3	17.2	Brittle
Brace anchor at 90°	12	16.3	2.12	271.7	155.2	30.3	Brittle
Coach screw 12 mm diameter	13	21.5	15.0	195.5	2.9	1.7	Ductile
Coach screw 16 mm diameter	14	34.2	15.0	88.3	21.4	2.7	Ductile
Framing bracket	15	16.8	15.0	27.8	9.6	4.7	Ductile
Inverted framing bracket	16	15.8	12.29	37.8	13.1	7.1	Ductile
SFS screws	17	18.5	2.52	14.4	12.7	11.9	Brittle

the test for very large deformations (around 40 mm—see Fig. 8a).

The performance of steel brace anchors and framing brackets was slightly better in terms of strength than the unreinforced concrete plugs, but markedly worse than the reinforced plugs in terms of both stiffness and strength.

4.3 Screws

Specimen 13 with a 12 mm diameter coach screw exhibited little movement until about 9.3 kN when the timber began to fail in bearing and the screw began to bend. Axial tension in the screw shank probably provided additional resistance at large slip

Fig. 7 Specimens pulled apart after test to destruction (a) Specimen 1, (b) Specimen 3, (c) Specimen 6, (d) Specimen 7

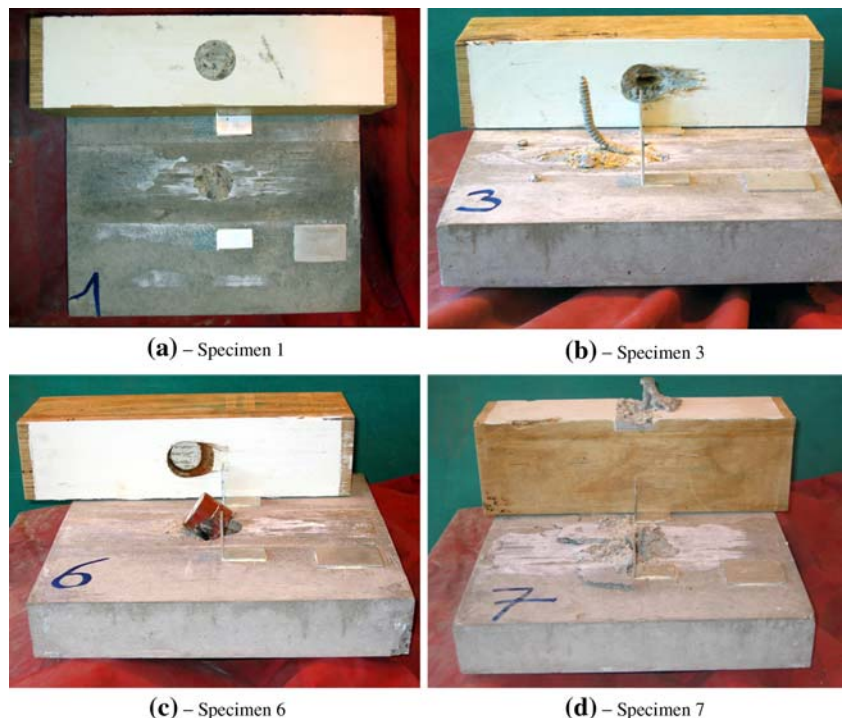
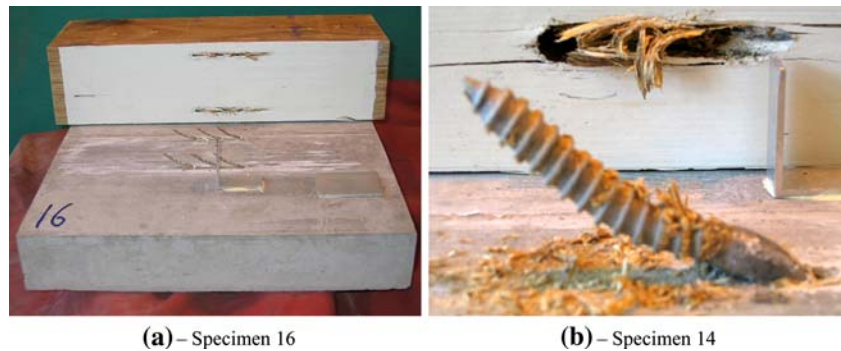


Fig. 8 Specimens pulled apart after test to destruction, at large displacements (40 and 60 mm, respectively) (a) Specimen 16, (b) Specimen 14



deformations, providing both direct tension and additional friction between the two surfaces.

The ultimate load of Specimen 14 (with a 16 mm diameter coach screw) was about 30% larger than that for the 12 mm diameter coach screw. The load capacity increased because the load was distributed into the wood over a larger area. In both cases the coach screw started to yield and the joint eventually failed due to crushing of the wood and concrete (see Fig. 8b). The larger diameter of the 16 mm diameter screw also caused the LVL to split longitudinally between veneers either side of the screw.

The overall performance of the specimens with coach screws was better than the specimens with steel brace anchor connectors, but it was worse than the performance of the specimens with reinforced concrete plugs.

The SFS screws used in Specimen 17 were specifically developed for connections between existing timber floors and concrete slabs. The two screws were driven at 45 and 135° to the loading direction to utilize the larger axial tensile stiffness of the fastener than the shear stiffness if the fastener only provides dowel action. The stiffness did not appear to be affected by the plywood layer placed between the concrete and LVL.

The screw in tension fractured during the test whereas the screw in compression broke into three parts. The brittle fracture of these screws was considered to be a very undesirable failure mode.

4.4 Discussion

The connection with the best mechanical performance was found to be the concrete plug reinforced with a screw, particularly the rectangular plug. This type of connection provided the greatest strength and

stiffness of all the specimens tested, due to the large bearing area provided by the plug. The post-peak performance and strength were both considerably enhanced by a large diameter coach screw. This detail can provide a very effective connection between the concrete slab and the LVL beam, resulting in an economical composite beam characterized by a high degree of composite action. The next most efficient connection was coach screws alone, with the advantage of ease of construction and no need to cut any notch. However, for a 12 mm diameter coach screw, the strength would be approximately doubled if used in conjunction with a circular plug and approximately quadrupled with a rectangular plug. The other connection systems are not considered to be useful in practise, the main reason being the poor mechanical performance which would lead to too many connectors to achieve adequate composite action between concrete and LVL.

The mechanical properties (average strength $F_{\max}$, average secant shear modulus at 40%, $k_{0.4}$ and 60%, $k_{0.6}$) reported in Table 5 can be employed for the design of the connection system between a concrete slab and a LVL joist. It has to be underlined, however, that with the small number of connectors tested, those properties can only indicate the relative performance of one connection system with respect to another. More specimens need to be tested in order to identify average and characteristic values representative of the actual performance of each connection system.

This investigation provided a qualitative assessment of a range of connection systems but a more comprehensive investigation needs to be conducted to provide design guidelines and design properties for the components and their dimensions [32]. Ideally, all of the materials should be close to failure at the

limiting slip to provide the best use of the materials. In particular, the bearing area of the concrete plug to LVL interface needs to be matched to the shear areas of both the concrete plug and the LVL to enhance the connector performance.

5 Performance of the connectors in the composite floor

The experimental test results were used to design the connection of a CCT floor, subject to the limitations discussed above. A floor span (L) of 8 m, representative of an office building, was chosen, with a 50 mm thick concrete slab poured over 20 mm thick plywood supported by 105×400 mm LVL joists spaced at 600 mm centres. The permanent load, including self-weight, was estimated as 2 kN/m^2 , and the imposed load was 3 kN/m^2 . The mechanical properties of the LVL and concrete were taken as those measured and assumed for the experimental investigation.

The maximum connector spacing was evaluated for the tested connection systems, using the design procedure proposed by Ceccotti et al. [33] based on Eurocode 5 [29, 30]. The flexibility of the connection was modelled using the γ -method by assuming the shear moduli of $k_{0.4}$ for serviceability and $k_{0.6}$ for ultimate limit state. The instantaneous deflection due to the imposed load and the long-term deflection due to the quasi-permanent load combination were limited to $L/300$, or 26.7 mm. The stresses in timber, concrete and connection systems were checked at the ultimate limit state for both the short- and long-term conditions. For this class of structure, the relative humidity of the surrounding air is only expected to exceed 85% for a few weeks per year, so the 2nd service class defined in Eurocode 5 was assigned so that the long-term creep coefficient for LVL was taken as 0.8 [29]. The creep coefficient of concrete was taken as 2.25. The creep coefficient for the connection systems was assumed to be twice as large as that for LVL, in accordance with Eurocode 5 [29]. The average values of connection strength ($F_{\max}$) reported in Tables 5 and 6 were reduced by 30% to estimate the characteristic values required for the design. The 30% reduction value is a very conservative assumption based on the outcomes of experimental push-out tests performed by other authors on a

larger number of specimens (see for example Steinberg et al. [10]).

Table 6 reports the outcomes of the design for the different connection systems. The maximum connectors spacing which satisfies all short- and long-term criteria was used as the only design variable, with equally-spaced connectors along the beam. The connection shear strength was found to be the most critical design property for almost all of the connection systems. Every other material was below its ultimate strength and the serviceability limit state of maximum deflection (Table 6) was not approached by any of the systems. The slips at the support at ULS are also reported in Table 6 for immediate comparison with the slip corresponding to strength measured in the shear tests (Table 5). Apart from the 12 mm diameter coach screw, all other connection systems exhibited slips less than 1 mm, so that the slip demand was always well below the slip at failure.

As expected, the screw reinforced rectangular concrete plug was the most effective, with a maximum spacing of 500 mm corresponding to 16 connectors equally-spaced along the length of the beam. If the 12 mm coach screw was used without a concrete plug, the maximum spacing would reduce to 325 mm (25 connectors).

The number of connectors needs to be minimised to make the CCT floor a viable and cost-effective solution. The strength of the reinforced plug system could most likely be improved by increasing the depth of the plug and the diameter of the screw so as to increase the strength and reduce the number of connectors that need to be cut into the beam. An improved connector design method and more exact inelastic analysis are part of an ongoing project.

6 Conclusions

An experimental program was carried out at the University of Canterbury to assess a range of connection systems that could be used to provide composite action between a concrete slab and its LVL support beams. The experiments were used to identify the most efficient and cost-effective connection systems for medium to long span composite floors. Asymmetric shear tests were used to measure the stiffness, strength and ductility of connector systems. Connector specimens included round and

Table 6 Strength and type of failure of the connectors, maximum connector spacing, deflection at SLS and slip at support at ULS in an 8-m long composite floor using different connection systems

Type of connector	Spec.No.	$F_{\max}$ (kN)	Type of failure	Maximum connector spacing (mm)	Deflection of SLS (mm)	Slip at support at ULS (mm)
Rectangular plug with screw	7 & 8	54.9	Ductile	500	9.6	0.170
Coach screw 16 mm diameter	14	34.2	Ductile	380	10.5	0.748
Coach screw 12 mm diameter	13	21.5	Ductile	325	9.6	3.363
Round plug with pipe	5 & 6	32.6	Ductile	320	10.7	0.469
Round plug with screw	3 & 4	31.4	Ductile	300	10.0	0.327
Brace anchor at 45°	11	19.3	Brittle	185	9.9	0.350
Brace anchor with rods	9	15.3	Brittle	135	9.4	0.078
Brace anchor	10	16.6	Brittle	145	9.4	0.070
Brace anchor at 90°	12	16.3	Brittle	145	9.2	0.062
Framing bracket	15	16.8	Ductile	190	11.3	0.975
Inverted framing bracket	16	15.8	Ductile	165	10.5	0.663
Round plug without screw	1 & 2	13.2	Brittle	120	9.6	0.177

rectangular concrete plugs, unreinforced or reinforced with screws or steel pipes, screwed connectors and folded sheet steel connectors that were nailed or screwed to the LVL.

The principal observations from the experimental tests were:

- The concrete plugs reinforced with a screw or steel pipe performed best, providing the best stiffness, strength, and post-peak behaviour. The concrete plug produced significant strength and stiffness, due to bearing at the interface between the LVL and concrete plug. The screw reinforcement enhanced the strength and post-peak behaviour, by acting as a tensile anchor and by providing additional dowel action. Replacing the screw with a steel pipe around the concrete plug reduced the joint strength.
- The strength and stiffness of these connection systems were found to depend upon the bearing area, so a rectangular plug provided better performance than a smaller round plug. The size of the screw should be increased for larger plugs in order to improve the post-peak behaviour.
- Sheet steel anchors screwed to the surface of the LVL produced lower strength, stiffness and ductility than that provided by specimens with concrete plugs. The orientation of the anchors had no effect on the connector performance which was governed by the connection between the anchor and the LVL.
- The coach screws specimens provided large initial stiffness and strength, however the peak strength was achieved at excessively large deformations (22 mm) due to the bending of the screw and crushing of timber at the interface. This large deformation would prevent them from developing their full strength in a flooring system. The SFS screws were significantly worse than the coach screws as they failed in a brittle manner.
- The rectangular concrete plug reinforced with a coach screw was shown to be capable of providing adequate composite action for an 8 m span floor system when installed at 500 mm spacing along the beam.

Further research will be performed on the rectangular concrete plugs reinforced with coach screws to optimise their dimensions and to increase their strength and stiffness. Tests will be performed on larger quantities of reduced scale shear specimens in order to evaluate the statistical values. Full-scale composite beams and floors with this connection will be tested in short- and long-term tests.

Acknowledgements The technical support and financial contribution provided by Carter Holt Harvey is gratefully acknowledged, with a special thanks to Mr. Hank Bier for the personal interest showed during the whole project. Particular thanks are also due to Ms. Eva Seibold for the experimental work and results presented in this paper. The indirect support provided by the New Zealand Earthquake Commission (EQC) for the first author is also gratefully acknowledged.



References

1. Ceccotti A (1995) Timber–concrete composite structures In: Blass HJ et al (eds) Timber engineering, Step 2. 1st edn Centrum Hout, The Netherlands
2. Ceccotti A, Fragiaco M, Giordano S (2007) Long-term and collapse tests on a timber–concrete composite beam with glued-in connection. *Mater Struct Special Volume Research for Reliable Timber Structures* 40(1): 15–25
3. Frangi A, Fontana M (2003) Elasto-plastic model for timber–concrete composite beams with ductile connection. *IABSE Struct Eng Int* 1:47–57
4. Piazza M, Ballerini M (2000) Experimental and numerical results on timber–concrete composite floors with different connection systems. In: 6th world conference on timber engineering WCTE 2000, Whistler Resort, British Columbia, Canada, 31 July–3 August
5. Fragiaco M, Amadio C, Macorini L (2006) Short- and long-term performance of the Tecnia stud connector for timber–concrete composite beams. *Mater Struct* Published Online
6. Gutkowsky RM, Brown K, Shigidi A, Natterer J (2004) Investigation of notched composite wood–concrete connections. *J Struct Eng* 130(10):1553–1561
7. Kuhlmann U, Michelfelder B (2004) Grooves as shear-connectors in timber–concrete composite structures. In: 8th world conference on timber engineering WCTE 2004, Lahti, Finland, 14–17 June 2004, vol 1, pp 301–306
8. Capozucca R (1998) Bond stress system of composite concrete–timber beams. *Mater Struct* 31(213):634–640
9. Clouston P, Bathon LA, Schreyer A (2005) Shear and bending performance of a novel wood–concrete composite system. *J Struct Eng* 131(9):1404–1412
10. Steinberg E, Selle R, Faust T (2003) Connectors for timber–lightweight concrete composite structures. *J Struct Eng* 129(11):1538–1545
11. Mascia NT, Soriano J (2004) Benefits of timber–concrete composite action in rural bridges. *Mater Struct* 37(266):122–128
12. Gelfi P, Giuriani E, Marini A (2002) Stud shear connection design for composite concrete slab and wood beams. *J Struct Eng* 128(12):1544–1550
13. Negrão JH, Oliveira FM, Oliveira CL (2006) Investigation on timber–concrete glued composites. In: 9th world conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
14. Lukaszewska E, Johnsson H, Stehn L (2006) Connections for prefabricated timber–concrete composite systems. In: 9th world conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
15. Dias A, Van de Kuilen J, Cruz H, Lopes S (2006) Densified veneer wood for notched joints in timber concrete composite structures. In: 9th world conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
16. Jorge L, Lopes S, Cruz H (2006) Castellated surface for timber–concrete composite connections. In: 9th world conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
17. Flach M, Schönbruch F (2006) Prefabricated wood–concrete slabs. In: 9th world conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
18. SFS (1999) SFS Verbundsystem VB Holz Beton Verbund (SFS connection system for timber–concrete composite structures) Published by SFS Unimarket AG 1999, PS50099D
19. Standards Association of New Zealand (1986) NZS 3112: method of test for concrete. Part 2: tests relating to the determination of strength of concrete. Wellington, New Zealand
20. Ranta-Maunus A (1995) Laminated veneer lumber and other structural sections In: Blass HJ et al (eds) Timber engineering, Step 1, 1st edn. Centrum Hout, The Netherlands
21. Futurebuild (2002) Span tables for residential building Published by Carter Holt Harvey, October 2002, New Zealand
22. Hsu LCY, Chauhan SS, King N, Lindstroem H (2003) Modulus of elasticity of stemwood vs. branchwood in 7-year-old *pinus radiata* families. *NZ J For Sci* 33(1):35–46
23. Werner H (1992) Holz-Beton-Verbunddecke mit einer neuartigen Fugenausbildung (Timber–concrete composite slab with a new type of connection system) Reprint from *Bauen mit Holz*, Reproduction from booklet, 4/92
24. Benitez MF (2000) Development and testing of timber–concrete shear connectors. In: 6th World Conference on Timber Engineering WCTE 2000, Whistler Resort, British Columbia, Canada, 31 July–3 August, paper No 832, CD
25. Kuhlmann U, Schänzlin J (2001) Grooves as shear connectors for timber–concrete composites decks. In: International RILEM Symposium on Joints in Timber Structures. Stuttgart, Germany, pp 283–290, 12–14 September 2001
26. Gerber C, Quast U, Steffens R (1993) Balkenschuhe als Verbundmittel für Holzbalkendecken mit mittragender Stahlbetondecke (Joist hangers as shear connectors for timber floors with activated reinforced concrete slab). *Beton Stahlbetonbau* 88(9):237–240
27. Seibold E (2004) Feasibility study for composite concrete/timber floor systems using laminated veneer lumber in New Zealand. Research report. University of Canterbury, New Zealand
28. Comité Européen de Normalisation (1991) Timber structures—Joints made with mechanical fasteners—General principles for the determination of strength and deformation characteristics, EN 26891. Bruxelles, Belgium
29. Comité Européen de Normalisation (2003) Eurocode 5—Design of timber structures—Part 1–1: General rules and rules for buildings, prEN 1995-1-1. Bruxelles, Belgium
30. Comité Européen de Normalisation (1996) Eurocode 5—Design of timber structures—Part 2: Bridges, ENV 1995-2. Bruxelles, Belgium
31. Comité Européen de Normalisation (2004) Eurocode 8—Design of structures for earthquake resistance—Part 3: Assessment and retrofitting of buildings Draft No 7 Stage 49. Bruxelles, Belgium
32. Kuhlmann U, Michelfelder B (2006) Optimised design of grooves in timber–concrete composite slabs. In: 9th world

- conference on timber engineering WCTE 2006, Portland, Oregon, USA, 6–10 August 2006, CD
33. Ceccotti A, Fragiaco M, Gutkowski RM (2002) Design of timber–concrete composite structures according to EC5–2002 version. In: Meeting thirty-five of the working commission W18-timber structures. CIB, international council for research and innovation, Kyoto (Japan), September, 10 pp